TEXORA · LECTURE SHEET

Introduction to Quality and Total Quality Management

What quality means and how its management evolved — the definitions, Garvin's dimensions of product and service quality, the TQM framework, the contributions of the quality gurus, the barriers to TQM, and the cost of quality.

TE-4108 · QUALITY ASSURANCE AND MANAGEMENT IN TEXTILES · TOPIC 1 · LEVEL 4 TERM 1

1. Need for quality and the evolution of quality

A textile firm sells into a chain of demanding judges: the spinner's yarn is judged by the weaver, the fabric by the dye house and the buyer, the garment by a retail customer who can switch brands at no cost. **Quality** is needed because defects are expensive in both directions — they cost money inside the factory (rework, seconds, claims) and they cost market outside it (lost buyers, price pressure, reputation). Quality is therefore not the inspection department's hobby; it is the firm's licence to trade.

The management of quality evolved in identifiable stages:

STAGE	APPROACH	RESPONSIBILITY	LIMITATION THAT FORCED THE NEXT STAGE
Inspection	Sort good from bad after production	Inspectors	Detects defects but never prevents them; cost of failure already incurred
Quality control (QC)	Statistical control of processes (control charts, sampling)	QC department	Controls processes but not the system that designs and plans them
Quality assurance (QA)	Planned system of procedures to build confidence (ISO 9000 family)	Management system, audited	Can degenerate into paperwork if improvement is not its engine

TQM / continuous improvement	Quality as everyone's job, customer-focused, fact-based improvement culture	Every employee, led from the top	Demands leadership persistence; fails as a slogan

2. Definitions of quality

- **Juran:** quality is *fitness for use* — the product does what the user needs, in the user's conditions.
- **Crosby:** quality is *conformance to requirements* — the specification, clearly stated, is the definition; "zero defects" is the standard.
- **Deming:** quality is aimed at the *needs of the consumer, present and future* — a moving target requiring continuous improvement of the system.
- **ISO 9000:** quality is the *degree to which a set of inherent characteristics fulfils requirements*.

The definitions complement rather than compete: a fabric can conform to its specification (Crosby) yet be unfit for the garment it enters (Juran) if the specification itself was wrong — which is why TQM insists the specification be derived from the customer. In textile terms: GSM within ±3 % is conformance; a fabric that sews without puckering at the buyer's line is fitness for use.

3. Dimensions of quality

Garvin's eight dimensions let a firm discuss product quality precisely instead of arguing about the word:

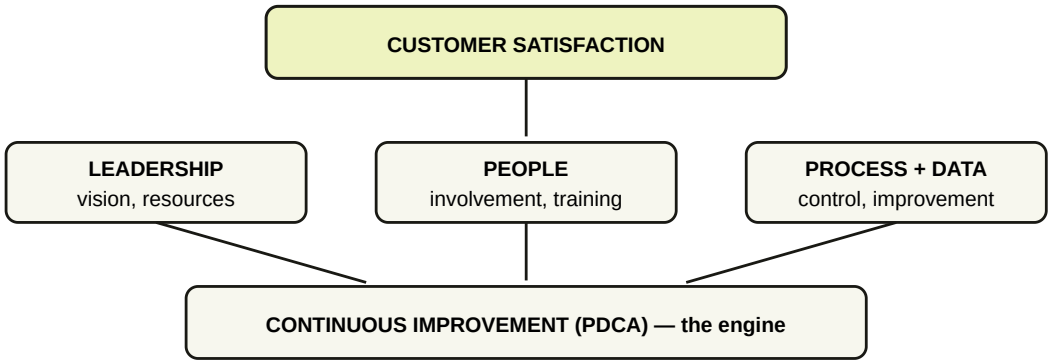
DIMENSION	MEANING	TEXTILE ILLUSTRATION
Performance	Primary operating characteristics	Fabric strength, yarn tenacity, garment warmth
Features	Secondary characteristics that supplement	Moisture management, anti-odour finish, extra pockets
Reliability	Probability of failure-free working life	Colour fastness holding over 50 washes
Conformance	Match to specification	GSM, shade (ΔE), measurements within tolerance
Durability	Useful life before deterioration	Pilling resistance, abrasion life of a fabric
Serviceability	Ease and quality of repair/after-service	Spare buttons supplied; claims handled quickly
Aesthetics	Look, feel, handle, smell	Fabric handle and drape; garment appearance

Perceived quality	Reputation-based judgement	Buyer's trust in a mill's consistency across seasons

Service quality has its own dimensions — reliability of promises, responsiveness, assurance (competence and courtesy), empathy, and tangibles (documents, samples, lab reports). An export mill sells service quality continuously: on-time shipment documents are as much "quality" to a buyer as the cloth.

4. Basic concepts and framework of TQM

Total Quality Management is a management approach centred on quality, based on the participation of all members, aiming at long-term success through customer satisfaction and benefits to all stakeholders. Its working concepts: **customer focus** (internal and external customers); **total employee involvement**; **process approach** (manage processes, not just results); **continuous improvement** (PDCA, Topic 2); **fact-based decisions** (data over opinion); **supplier partnership**; and **leadership commitment** as the enabling condition of all the rest.



Remove any block and the structure leans on slogans: leadership without data is exhortation; data without people is filing.

Fig. 1 — The TQM framework: customer satisfaction as the roof, held up by leadership, people and process-with-data, driven by continuous improvement.

5. Contributions of the quality gurus

GURU	CORE CONTRIBUTION	TEXTILE ECHO
W. Edwards Deming	System thinking; 14 points for management; PDCA cycle; "drive out fear"; variation as the enemy	Most defects are system-caused (materials, machines, methods) — not operator-caused
Joseph M. Juran	Fitness for use; the Juran trilogy (quality planning, control, improvement); Pareto principle in quality	Plan quality from the customer's use backwards

Philip B. Crosby	"Quality is free"; zero defects; do it right first time; price of nonconformance	Rework and seconds cost more than prevention
Kaoru Ishikawa	Cause-and-effect (fishbone) diagram; quality circles; company-wide quality control	Front-line teams solving their own section's defects
Genichi Taguchi	Quality loss function; robust design — quality built in by design, not inspection (Topic 4)	Being "within tolerance" is not the same as being on target
Armand V. Feigenbaum	"Total quality control" — quality as a company-wide, integrated system	The direct ancestor of TQM's "total"

6. Barriers to TQM

TQM programmes fail for organisational, not technical, reasons: lack of visible top-management commitment (quality delegated to a department); treating TQM as a campaign with an end date; no training, or training without changed systems to apply it in; measurement systems that reward output volume while preaching quality; fear — operators hiding defects because defects bring punishment; supplier relationships kept adversarial on price alone; and impatience — abandoning the approach when the first quarter's results lag the slogans. Each barrier maps to a framework block in Fig. 1: TQM fails at whichever block was faked.

7. Cost of quality

The **cost of quality (CoQ)** makes quality financial. It has four categories: **prevention** (training, process design, supplier evaluation — spent to stop defects occurring), **appraisal** (inspection, testing, audits — spent to detect), **internal failure** (rework, seconds, reprocessing, downtime — defects caught before delivery), and **external failure** (claims, returns, discounts, lost buyers — defects caught by the customer). Crosby's point: failure costs are usually the largest and the least measured.

> **WORKED EXAMPLE — COST OF QUALITY OF A DYEING AND FINISHING MILL (ONE YEAR)**
>
> Prevention Tk 2.5 lakh (13.9 % of CoQ); Appraisal Tk 4.0 lakh (22.2 %); Internal failure Tk 6.5 lakh (36.1 %); External failure Tk 5.0 lakh (27.8 %).
>
> Total CoQ = 2.5 + 4.0 + 6.5 + 5.0 = **Tk 18.0 lakh** = 3.0 % of the mill's Tk 6 crore sales.
>
> Failure costs (internal + external) = Tk 11.5 lakh = **63.9 % of all quality cost**.
>
> Management reading: the mill spends nearly two-thirds of its quality money on the consequences of defects. The TQM investment case writes itself — shifting even part of the failure total into prevention (which typically costs a fraction of the failure it removes) reduces total CoQ while

> improving delivery reliability. A mill that cannot produce this table does not know what quality costs it; it only knows what the QC department's salaries are.

Glossary

Quality — fitness for use / conformance to requirements / degree to which characteristics fulfil requirements (definitions in Section 2). **QC** — operational techniques to control quality. **QA** — the planned system of activities giving confidence that quality will be achieved. **TQM** — organisation-wide, customer-focused continuous improvement of quality. **Cost of quality** — prevention + appraisal + internal failure + external failure costs. **Zero defects** — Crosby's standard: the performance goal is no defects, not an "acceptable" number.

Key points

• Quality management evolved: inspection → QC → QA → TQM; each stage exists because the previous one could not prevent defects system-wide.
• Definitions converge: derive the specification from the customer's use, then conform to it.
• Garvin's eight dimensions make product quality discussable; service quality has its own five.
• Gurus: Deming (system, PDCA), Juran (fitness for use, trilogy), Crosby (zero defects, quality is free), Ishikawa (fishbone, circles), Taguchi (loss function), Feigenbaum (total quality control).
• Worked CoQ: Tk 18.0 lakh total, failure costs 63.9 % — the financial argument for prevention.

Practice

1. A mill's annual costs: prevention Tk 3.0 lakh, appraisal Tk 5.0 lakh, internal failure Tk 9.0 lakh, external failure Tk 3.0 lakh, on sales of Tk 8 crore. Compute total CoQ, its % of sales, and the failure share.
2. A fabric passes every laboratory specification yet the garment maker rejects it for sewing puckering. Analyse with Juran and Crosby definitions: where did quality fail?
3. Match each guru to the idea: zero defects; fitness for use; fishbone diagram; loss function; 14 points.
4. Give three barriers to TQM in a garment factory and the framework block each one breaks.

Answer key

1. Total CoQ = 3 + 5 + 9 + 3 = Tk 20.0 lakh. % of sales = 20 ÷ 800 × 100 = 2.5 %. Failure = internal + external = 12 lakh = 60 % of CoQ.
2. By Crosby's definition quality was achieved — the product conformed to its requirements. By Juran's it failed: the fabric was not fit for the customer's actual use. The failure is therefore in the

requirements themselves — the specification omitted a sewability characteristic that mattered (e.g. seam slippage/friction behaviour) — and the corrective action is in quality planning: derive specifications from the customer's process, not from the laboratory's convenience. Inspection cannot catch a wrong specification.

3. Zero defects — Crosby. Fitness for use — Juran. Fishbone (cause-and-effect) diagram — Ishikawa. Loss function — Taguchi. 14 points — Deming.
4. Examples: (i) Management announces TQM but keeps rewarding only shipment volume — breaks **leadership** (the stated and rewarded systems conflict). (ii) Operators hide defects for fear of punishment — breaks **people/involvement**; without honest defect data the process block also starves. (iii) No defect data are collected by operation, so improvements are argued from opinion — breaks **process + data**. (Also acceptable: adversarial supplier squeezing breaks partnership; campaign mentality breaks continuous improvement.)

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